

Video-Text Multi-Instance Retrieval in Egocentric Video: From a JPoSE Baseline to AVION ViT-L Fine-Tuned with Symmetric Multi-Similarity Loss on EPIC-KITCHENS-100

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Abstract

We address the EPIC-KITCHENS-100 Multi-Instance Retrieval (MIR) challenge through three progressively stronger approaches: (i) a JPoSE-based zero-fine-tuning baseline that exploits part-of-speech embeddings; (ii) a score-level ensemble of three pre-trained retrieval models (JPoSE, MI-MM and MMEN) with optional graph-diffusion re-ranking on visual and textual k -nearest-neighbour graphs; and (iii) end-to-end fine-tuning of an AVION ViT-L/14 dual encoder with the Symmetric Multi-Similarity (SMS) loss, which directly consumes the soft-label relevancy matrices, enhanced with horizontal-flip and longer-clip test-time augmentation. Across six iterations, average nDCG on the official Codabench leaderboard rises from 53.53 (JPoSE alone) to 55.82 (ensemble + re-ranking) and finally to 69.68 (AVION ViT-L + SMS, 16 epochs, with TTA), an absolute gain of +16.15 nDCG. We analyse where each gain originates, whether encoder capacity, soft-label supervision, test-time augmentation and additional epochs, and outline a model-soup-style ensemble of differently-trained ViT-L+SMS instances as future work, motivated by the compute and offline-evaluation constraints encountered. Our code will be released at: <https://github.com/ISIS-4825-Assignments/Multi-Instance-Retrieval-EK-100>

viewpoint, illumination and motion variability. Models must therefore interpret actions and contextual relationships at fine granularity in order to establish precise correspondences between visual content and textual descriptions.

The reference resource for this problem is the EPIC-KITCHENS-100 dataset [2], which contains over 100 h of egocentric kitchen video and approximately 90,000 narrated action segments. The associated Multi-Instance Retrieval (MIR) challenge formalises the task: given a textual query, the system must rank 9,668 trimmed video segments from the test split (and vice versa) by their semantic relevance, where ground-truth relevance is provided as a continuous soft-label matrix rather than as one-to-one binary correspondences [2, 16]. This formulation acknowledges that several segments may match a query at different degrees of similarity and so, is evaluated with order-sensitive metrics like normalized discounted cumulative gain (nDCG) [7] and mean average precision (mAP) rather than with the conventional Recall@K used in third-person retrieval. Despite recent progress in multimodal foundation models, semantic alignment between egocentric video and natural language remains difficult due to dynamic perspective changes, occlusions and multi-object interactions hinder the extraction of robust visual representations, and the fact that most existing losses do not exploit the graded relevance signal provided by the challenge.

1. Introduction

The growth of audiovisual content has driven the development of methods capable of analysing, indexing and automatically retrieving visual information. Within this landscape, multimodal video-text retrieval has consolidated as a relevant area at the intersection of computer vision and natural language processing. An especially challenging scenario is that of *egocentric video*, captured from a first-person perspective using head- or body-mounted cameras. Such recordings register natural interactions between hands, objects and the surrounding environment, introducing strong

2. Methodology

We approached EK-100 MIR as an iterative search through the design space, starting from a feature-based zero-fine-tuning baseline and ending with end-to-end fine-tuning of a strong egocentric backbone with soft-label supervision. The three approaches share a common formalization, stated once before each iteration is described.

2.1. Task formalization and common notation

Let $\mathcal{V} = \{v_i\}_{i=1}^{N_v}$ denote the $N_v = 9,668$ test video segments and $\mathcal{T} = \{t_j\}_{j=1}^{N_t}$ the $N_t = 3,842$ unique text queries

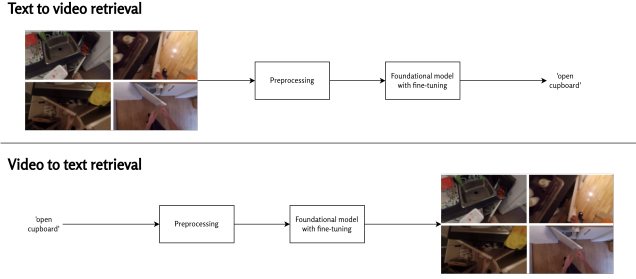


Figure 1. Overview of the multi-instance retrieval pipeline shared by the three approaches: visual and textual encoders project segments and queries into a shared space, and the resulting similarity matrix S is ranked and evaluated against the soft-label matrix R . Image components adapted from [2].

on the test split. A retrieval system produces a similarity matrix $S \in \mathbb{R}^{N_v \times N_t}$ such that S_{ij} is large when v_i is semantically relevant to t_j . Ground-truth relevance is encoded in the soft-label matrix $R \in [0, 1]^{N_v \times N_t}$ [16] and the official challenge metric is the average normalized discounted cumulative gain

$$\text{nDCG}_{\text{AVG}} = \frac{1}{2}(\text{nDCG}_{V \rightarrow T}(S, R) + \text{nDCG}_{T \rightarrow V}(S^T, R^T)) \quad (1)$$

which captures both retrieval directions on equal footing. The three approaches differ only in how S is constructed while the evaluation protocol is identical across them. Figure 1 summarizes the resulting pipeline.

2.2. Approach 1: JPoSE Part-of-Speech baseline

Our entry point is a feature-based, zero-fine-tuning baseline using JPoSE [15]. It projects segments and queries into verb, noun and full-action sub-spaces, learned jointly with triplet supervision. Let $\phi_V^v, \phi_N^v, \phi_A^v$ and $\phi_V^t, \phi_N^t, \phi_A^t$ denote the visual and textual sub-encoders. We compute joint embeddings as the concatenation $\mathbf{f}_v(v_i) = [\phi_V^v(v_i); \phi_N^v(v_i); \phi_A^v(v_i)]$ and analogously $\mathbf{f}_t(t_j)$, and obtain $S_{ij}^{\text{JPoSE}} = \cos(\mathbf{f}_v(v_i), \mathbf{f}_t(t_j))$. We use the released JPoSE.BEST checkpoint without modification on pre-extracted features. By design this approach cannot exploit R and inherits the limitations of triplet supervision with binary positives and negatives, but it provides a reproducible reference point against which all subsequent iterations are compared.

2.3. Approach 2: score-level ensemble with graph-diffusion re-ranking

The second approach asks how far a purely inference-time strategy can carry the JPoSE baseline by combining several pre-trained retrieval models and refining the result with neighborhood-aware re-ranking. Three complemen-

tary models are queried independently: S^{JPoSE} as in Section 2.2; $S^{\text{MI-MM}}$, the multi-instance MaxMargin model of Miech et al. [10] with S3D-HowTo100M features at the released best epoch (202); and S^{MMEN} , an MLP-based Multi-Modal Embedding Network released as a baseline in [15].

Step 1: row-wise normalization and equal-weight fusion. The three matrices live on different scales (cosine similarities for JPoSE/MMEN, max-margin distances for MI-MM), so a direct sum is biased towards the highest-magnitude model. We min-max normalize each matrix *row-wise* so that, for every video query, scores against the 3,842 candidates are mapped to $[0, 1]$:

$$\tilde{S}_{ij}^{(m)} = \frac{S_{ij}^{(m)} - \min_k S_{ik}^{(m)}}{\max_k S_{ik}^{(m)} - \min_k S_{ik}^{(m)}} \quad (2)$$

$$S^{\text{ens}} = \frac{1}{3}(\tilde{S}^{\text{JPoSE}} + \tilde{S}^{\text{MI-MM}} + \tilde{S}^{\text{MMEN}}) \quad (3)$$

Equal weights are the natural default in the absence of a relevance-aligned validation signal.

Step 2: graph-diffusion re-ranking on visual and textual k-NN graphs. Inspired by diffusion-based re-ranking in image retrieval [6] and k-reciprocal encoding [19], we propagate ensemble scores through the local geometry of the retrieval space in both modalities. After L2-normalising each row of S^{ens} , a sparse visual affinity matrix $A_{vv} \in \mathbb{R}^{N_v \times N_v}$ stores cosine similarity between the score profiles of each pair of segments if either is among the top- k visual neighbors of the other. The same construction on $(S^{\text{ens}})^T$ yields a textual affinity A_{tt} . We then expand the score matrix in both directions and blend with the original ensemble:

$$S^{\text{exp}} = \frac{1}{2}(A_{vv} S^{\text{ens}} + (A_{tt} (S^{\text{ens}})^T)^T), \quad (4)$$

$$S^{\text{rerank}} = (1 - \alpha) S^{\text{ens}} + \alpha S^{\text{exp}} \quad (5)$$

with default $k = 20$, $\alpha = 0.3$. Both the ensemble matrix (3) and the re-ranked matrix (5) are submitted to Codabench as separate iterations, isolating the contribution of re-ranking.

2.4. Approach 3: AVION ViT-L fine-tuned with the SMS loss

The third approach replaces inference-time fusion with end-to-end fine-tuning of a strong CLIP-style egocentric backbone, supervised with the soft-label relevance matrix. The visual encoder is the AVION ViT-L/14 of Zhao and Krähenbühl [17], pre-trained on Ego4D via LaViLa [18] and the textual encoder is the CLIP text transformer paired with that AVION checkpoint. Both encoders are fine-tuned jointly with the Symmetric Multi-Similarity loss of Wang, Wang and Chau [13].

Symmetric Multi-Similarity loss. Within a training batch of B video-text pairs, let $\mathbf{x}_i, \mathbf{y}_i \in \mathbb{R}^d$ denote the L2-normalized visual and textual embeddings of pair i , $r_{ij} \in [0, 1]$ the soft relevance between v_i and t_j extracted from R [16], and $s_{ij} = \mathbf{x}_i^\top \mathbf{y}_j$. The SMS loss extends the multi-similarity formulation of [11] so that, instead of binary pair weights, the contribution of each pair is modulated by r_{ij} . With margin θ , relevance threshold τ and temperatures β_p, β_n ,

$$\mathcal{L}_i^{V \rightarrow T} = \frac{1}{\beta_p} \log \left(1 + \sum_{j: r_{ij} \geq \tau} r_{ij} e^{-\beta_p (s_{ij} - \theta)} \right) + \frac{1}{\beta_n} \log \left(1 + \sum_{j: r_{ij} < \tau} (1 - r_{ij}) e^{\beta_n (s_{ij} - \theta)} \right) \quad (6)$$

and the full loss is the symmetric counterpart $\mathcal{L} = \frac{1}{2}(\mathcal{L}^{V \rightarrow T} + \mathcal{L}^{T \rightarrow V})$ averaged over the batch. Compared with InfoNCE-like contrastive losses, SMS pulls together pairs in proportion to their relevance and pushes apart pairs in proportion to $1 - r_{ij}$, providing a supervision signal directly aligned with the nDCG metric.

Fine-tuning configuration. Fine-tuning is launched from the AVION pretrain checkpoint via `amplus.finetune.py` from a patched fork of the public SMS-Loss codebase [12] on a single A100. Key flags are: `--model CLIP_ViT_L14`, `--batch-size 48`, `--lr 2e-5`, `--loss-margin 0.6`, `--loss-thres 0.1`, with `--use-fast-conv1`, `--grad-checkpointing` (essential for ViT-L on a single GPU) and `--use-flash-attn` when available. Here, FlashAttention [3] roughly halves attention memory at no accuracy cost. We use AdamW for optimization [9] and one numbered checkpoint is saved per epoch. The relevance threshold $\tau = 0.1$ excludes pairs whose soft relevance is too small to provide a useful gradient and the margin $\theta = 0.6$ controls the separation enforced between positive and negative pairs; these hyperparameters follow the public SMS-Loss configuration of [13]. The pipeline supports resuming a training run from any saved checkpoint by passing `--pretrain-model` and `--resume` simultaneously. We use this hook to perform an initial 10-epoch run, followed by an additional 6 epochs of fine-tuning resumed from the epoch-10 checkpoint, for a total of 16 epochs.

Divergences from the published recipe. Two configuration choices depart from the SMS-Loss recipe of Wang et al. [13] and shape the upper bound of what our pipeline can achieve. First, our effective batch size of 48 on a single A100 is approximately five times smaller than the 240 samples (60 per GPU $\times$ 4 GPUs) reported for ViT-L by the orig-

inal authors. Smaller batches reduce the diversity of negatives visible to the SMS denominator in Eq. (6) and weaken the metric-learning signal. Second, our base run is restricted to 10 epochs (extended to 16 in iteration 6) against the 100 epochs of the published recipe. Both deviations are quantified in Section 6 and motivate the future-work direction in Section 7.

Inference and test-time augmentation. Inference is performed by `test.mir.py`, which encodes all test video segments and text queries and writes a similarity matrix in the same format as Approaches 1 and 2. We compare two settings: a *standard* pass with default 8-frame clips, and a *TTA* pass with `--flip --clip-length 32` that averages the embeddings of the original and horizontally-flipped clips and samples 32 frames per segment. Flip-averaging is the canonical video adaptation of [8] and exploits the near-invariance of kitchen activities to horizontal reflection; the longer clip length is motivated by the long tail of segment durations.

3. Results

3.1. Evaluation protocol

All iterations are evaluated under a single protocol on the official EPIC-KITCHENS-100 MIR benchmark on Codabench¹. Each submission is a Codabench-compatible ZIP containing the $9,668 \times 3,842$ similarity matrix serialized as a protocol-2 pickle with a numpy namespace shim. It is scored by the grader against the held-out test relevance matrix [16] and the grader returns the average nDCG (Eq. 1) used to rank submissions. Because the test relevance matrix is not publicly released, we cannot compute mAP or per-direction nDCG offline. Therefore, we report the official Codabench leaderboard nDCG AVG as the single primary metric throughout the paper. As a sanity check on the implementation of Approach 1, the JPoSE codebase exposes an internal train-set validation routine on which we obtained an offline nDCG of 0.690 and mAP of 0.734, matching the figures of Wray et al. [15] before any submission was made. These offline numbers are not directly comparable to the public Codabench score for the same submission (53.53 nDCG) and are therefore not used for cross-iteration comparison.

3.2. Comparative results across iterations

Table 1 reports the official Codabench nDCG AVG for each of the six iterations described in Sections 2.2–2.4, together with the marginal contribution of every design change introduced over the previous iteration. The trajectory closes more than half of the gap between feature-based baselines and the current public state of the art (CR-CLIP at

¹<https://www.codabench.org/competitions/12008/>

229 82.10 nDCG [5]) on a single GPU and using only sixteen
230 epochs of fine-tuning.

231 The results show a clear hierarchy of effects. First, en-
232 semble fusion of the three frozen pretrained models con-
233 tributes +1.78 nDCG over the JPoSE baseline, indicat-
234 ing that JPoSE, MI-MM and MMEN are complementary
235 enough for score averaging to recover variance that none
236 of them captures individually. Second, graph-diffusion re-
237 ranking adds +0.51 nDCG on top of the ensemble at no
238 training cost, consistent with the magnitude of the gains re-
239 ported in image retrieval [6]. Third, replacing the inference-
240 time pipeline with the AVION ViT-L + SMS configura-
241 tion produces the largest jump in the table, +12.98 nDCG.
242 Fourth, post-fine-tuning refinements behave as expected:
243 TTA adds +0.68 nDCG at no training cost, and continu-
244 ing fine-tuning for six additional epochs on top of the ini-
245 tial 10-epoch run (16 epochs in total) adds +0.20 nDCG,
246 suggesting that single-checkpoint training is approaching its
247 plateau.

248 4. Discussion

249 The per-iteration breakdown in Table 1 concentrates 80%
250 of the total improvement (12.98 of 16.15 nDCG points) in
251 the single transition from Approach 2 to Approach 3. That
252 is from a frozen-feature ensemble to an end-to-end fine-
253 tuned ViT-L driven by soft-label supervision. The asym-
254 metry along the iteration axis is itself the central observa-
255 tion, and it can be unpacked along three connected lines: the
256 low ceiling of inference-time refinements over frozen back-
257 bones, the dominant effect of the encoder-and-loss swap,
258 and the diminishing returns of post-fine-tuning refinements
259 that motivate the future-work direction.

260 Approach 2 fuses three pre-trained models that were
261 never optimized for the soft-label setting: JPoSE and
262 MMEN are triplet-supervised, MI-MM is binary multi-
263 instance MaxMargin, and none sees the relevance matrix R
264 during training. Equal-weight averaging of their normalized
265 scores recovers +1.78 nDCG on top of JPoSE, in line with
266 the canonical observation that retrieval ensembles reduce
267 variance without correcting bias [14]. Because the three
268 models share the same failure mode (they ignore graded
269 relevance), averaging them produces a more stable estimate
270 of the same flawed objective rather than one that is closer
271 to nDCG. Equal weights were used in the absence of a
272 relevance-aligned validation signal and weighted averaging
273 tuned on a held-out split could plausibly add a fraction of a
274 point but is unlikely to overcome the bias ceiling. Graph-
275 diffusion re-ranking adds another +0.51 nDCG at no train-
276 ing cost, consistent with magnitudes reported for diffusion
277 in image retrieval [6]. A small $k = 20$ keeps the affinity
278 graph sparse and avoids over-smoothing, moderate $\alpha = 0.3$
279 blends rather than replaces the original signal, and the sym-
280 metric construction in Eq. (5)—propagating through both

visual and textual graphs—prevents either modality from
dominating the result.

The single largest jump, +12.98 nDCG between itera-
tions 3 and 4, is driven by three coupled changes that we
deliberately bundled into a single iteration boundary by re-
placing pre-extracted features with an end-to-end ViT-L/14
encoder pre-trained on egocentric video [17, 18], replacing
binary triplet/MaxMargin objectives by the SMS loss [13]
that consumes the soft-label matrix R directly, and jointly
fine-tuning both encoders rather than freezing them. Dis-
entangling the relative contribution of each factor would re-
quire ablations beyond our compute budget. What is clear
is that the gain dominates everything else by an order of
magnitude. We attribute most of the effect to the combina-
tion of the strong egocentric backbone and the relevance-
aware loss. The SMS objective in Eq. (6) weights both pos-
itive and negative pairs by their soft relevance, which aligns
the gradient signal with the official nDCG metric (Eq. (1))
and is the only mechanism in our pipeline that directly opti-
mises it. The size of the gain is consistent with the observa-
tion in [16] that graded-relevance objectives can reshape re-
trieval orderings well beyond what binary contrastive losses
achieve.

Once the encoder has been fine-tuned end-to-end with
SMS, however, further gains diminish quickly. TTA
(--flip --clip-length 32) yields +0.68 nDCG
with no training cost. Flip-averaging adapts the canon-
ical recipe of [8] to video and exploits the near-symmetry
of kitchen scenes under horizontal reflection, while the
longer clip length addresses the long tail of segment du-
rations. Extending fine-tuning from 10 to 16 epochs adds
a further +0.20 nDCG, suggesting the model is approach-
ing the plateau of single-checkpoint training. This corre-
sponds precisely to the regime in which model-averaging
strategies tend to be most cost-effective and which moti-
vates the future-work direction in Section 7. The practi-
cal implication of Table 1 is that, under a tight compute
budget, encoder fine-tuning with relevance-aware supervi-
sion is the highest-yield investment by a wide margin: re-
ranking and TTA are essentially free and should be applied
by default, ensemble fusion of frozen baselines is also cheap
but quickly hits a ceiling, and longer training is subject
to rapidly diminishing returns. By contrast, the encoder
swap returned +12.98 nDCG for the cost of a single A100
day. This ordering is consistent with what both the SMS-
Loss authors [13] and the CR-CLIP authors [5] report on
the same benchmark: the largest contribution to their final
scores comes from the encoder choice and the relevance-
aware loss, with subsequent refinements closing a smaller
residual gap.

Table 1. Official Codabench nDCG AVG (%) across the six iterations of Section 2. Each row attributes a delta to the single design change introduced over the previous iteration. Best per approach in **bold**.

It.	Approach	Single change	nDCG AVG	Δ
1	A1 – JPoSE	Feature-based baseline (no fine-tune)	53.53	–
2	A2 – Ensemble	+ 3-model ensemble (no re-ranking)	55.31	+1.78
3	A2 – Ensemble	+ Graph-diffusion re-ranking ($k=20$, $\alpha=0.3$)	55.82	+0.51
4	A3 – AVION+SMS	Replace pipeline by AVION ViT-L + SMS, 10 ep	68.80	+12.98
5	A3 – AVION+SMS	+ TTA: <code>--flip --clip-length 32</code>	69.48	+0.68
6	A3 – AVION+SMS	+ Six additional fine-tuning epochs (10 $\rightarrow$ 16)	69.68	+0.20
<i>Public leaderboard SOTA, CR-CLIP [5]</i>			82.10	–
Total improvement, It. 1 $\rightarrow$ It. 6			–	+16.15

5. Conclusions

We studied the EPIC-KITCHENS-100 Multi-Instance Retrieval challenge through three configurations evaluated under a single Codabench protocol. Starting from a JPoSE feature-based baseline at 53.53 nDCG, a score-level ensemble of three frozen retrieval models combined with graph-diffusion re-ranking lifts performance to 55.82 nDCG, while an AVION ViT-L/14 dual encoder fine-tuned with the SMS loss and complemented by horizontal-flip plus 32-frame test-time augmentation reaches 69.68 nDCG over sixteen epochs of single-GPU training. The strongest configuration recovers more than half of the gap between feature-based baselines and the current public state of the art (CR-CLIP, 82.10 nDCG [5]) using a single A100 instance and the publicly released SMS-Loss training code. End-to-end fine-tuning with a relevance-aware loss concentrates the bulk of the achievable improvement on a single GPU, while inference-time strategies (ensembling, graph-diffusion re-ranking and test-time augmentation) contribute small but reliable deltas at zero training cost.

6. Limitations

The results are constrained by two factors. First, Approach 3 was evaluated under a limited training budget, which precluded larger effective batch sizes, multi-GPU training and multi-seed re-training. Several knobs that are routine in the EK-100 MIR literature were therefore left unexplored. These include the contribution of larger batches (which directly affects the SMS gradient through the implicit hard-negative mining in Eq. (6)), systematic sweeps over θ , τ , β_p , β_n , longer fine-tuning schedules above 16 epochs, and variance estimation through repeated runs with different random seeds. As a consequence, the bundled +12.98 nDCG between Configuration 3 and Configuration 4 cannot be ablated into the individual contributions of encoder, loss and end-to-end training within this study. Second, the EK-100 challenge releases the soft-label rele-

vance matrix only for the training split; the test relevance matrix required to compute nDCG and mAP offline is held out by the organizers and accessible exclusively through the Codabench grader. Model selection during fine-tuning therefore had to rely on training-loss trajectories rather than on a relevance-aligned validation signal, and every reported number in Table 1 required a Codabench submission, which capped the practical iteration budget.

7. Future Work

The most promising extension consistent both with the empirical evidence in Table 1 and with the SMS-Loss recipe of Wang et al. [13] is the *score-level ensemble of multiple SMS-supervised checkpoints* described in Section 3.3 of that paper; reproducing it exceeded the available single-GPU compute budget. Concretely, the published recipe combines six independent models: three Adaptive MI-MM variants with different learning-rate schedules and three SMS variants with different relaxation thresholds ($\tau \in \{0.05, 0.10, 0.12\}$). All six are trained for 100 epochs from the same AVION pretrain [17] and are combined by directly summing their test similarity matrices, achieving an additional +1.1 mAP and +0.6 nDCG over the best single model. Replicating this ensemble on top of our AVION+SMS pipeline is a natural and well-defined extension: the row-wise normalization and equal-weight averaging of Approach 2 (Eq. (3)) provide a drop-in replacement for the direct sum, and the graph-diffusion re-ranking of Eq. (5) can be stacked on top at no training cost. Three observations support this extension. First, longer training of a single ViT-L instance yields rapidly diminishing returns past 10–16 epochs, so additional compute is better invested in a few short complementary models than in extending a single one indefinitely. Second, model-soup-style strategies—in which several models fine-tuned with different hyperparameters but the same initialization are combined—consistently match or exceed the best individual model in image classification [14] and have already been shown to work in this exact retrieval setting [13]. Third, Approach 2 already validates the inference-time fu-

sion machinery on weak frozen baselines, so deploying it on stronger fine-tuned ViT-L+SMS instances is essentially mechanical. The main obstacle is scale: the published ensemble requires six independent ViT-L+SMS models trained for 100 epochs each, placing it beyond the experimental scope of this study. This direction is the natural continuation of the AVION+SMS pipeline, conditional on multi-GPU infrastructure, in line with the 4×4 RTX 6000 Ada configuration used by Wang et al. [13] for ViT-B and the 48 GB GPUs used for ViT-L. Beyond the ensemble direction, two further extensions follow from the limitations above: integrating the context-refinement mechanism of CR-CLIP [5] on top of our AVION+SMS pipeline (closing a documented portion of the gap to the public leaderboard), and replacing the SMS loss in Eq. (6) with a directly differentiable nDCG surrogate adapted to graded relevance (Smooth-AP [1] or a SoDeep-style ranker [4]).

References

- [1] Andrew Brown, Weidi Xie, Vicky Kalogeiton, and Andrew Zisserman. Smooth-AP: Smoothing the path towards large-scale image retrieval. In *ECCV*, pages 677–694, 2020. 6
- [2] Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Antonino Furnari, Jian Ma, Evangelos Kazakos, Davide Moltisanti, Jonathan Munro, Toby Perrett, Will Price, and Michael Wray. Rescaling egocentric vision: Collection, pipeline and challenges for EPIC-KITCHENS-100. *International Journal of Computer Vision*, 130(1):33–55, 2022. 1, 2
- [3] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. *Advances in Neural Information Processing Systems (NeurIPS)*, 35:16344–16359, 2022. 3
- [4] Martin Engilberge, Louis Chevallier, Patrick Pérez, and Matthieu Cord. SoDeep: A sorting deep net to learn ranking loss surrogates. In *CVPR*, pages 10784–10793, 2019. 6
- [5] Jing He, Yiqing Wang, Lingling Li, Kexin Zhang, and Puhua Chen. ContextRefine-CLIP for EPIC-KITCHENS-100 multi-instance retrieval challenge 2025. *arXiv preprint arXiv:2506.10550*, 2025. 4, 5, 6
- [6] Ahmet Iscen, Giorgos Tolias, Yannis Avrithis, Teddy Furon, and Ondřej Chum. Efficient diffusion on region manifolds: Recovering small objects with compact CNN representations. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2077–2086, 2017. 2, 4
- [7] Kalervo Järvelin and Jaana Kekäläinen. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446, 2002. 1
- [8] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. ImageNet classification with deep convolutional neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*, 25:1097–1105, 2012. 3, 4
- [9] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. 3
- [10] Antoine Miech, Jean-Baptiste Alayrac, Lucas Smaira, Ivan Laptev, Josef Sivic, and Andrew Zisserman. End-to-end learning of visual representations from uncurated instructional videos. In *CVPR*, pages 9879–9889, 2020. 2
- [11] Xun Wang, Xintong Han, Weilin Huang, Dengke Dong, and Matthew R. Scott. Multi-similarity loss with general pair weighting for deep metric learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5022–5030, 2019. 3
- [12] Xiaoqi Wang, Yi Wang, and Lap-Pui Chau. SMS-Loss: Code release for “symmetric multi-similarity loss for EPIC-KITCHENS-100 multi-instance retrieval challenge 2024”. <https://github.com/xqwang14/SMS-Loss>, 2024. Accessed: 2026-05-07. 3
- [13] Xiaoqi Wang, Yi Wang, and Lap-Pui Chau. Symmetric multi-similarity loss for epic-kitchens-100 multi-instance retrieval challenge 2024, 2024. 2, 3, 4, 5, 6
- [14] Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, pages 23965–23998, 2022. 4, 5
- [15] Michael Wray, Diane Larlus, Gabriela Csurka, and Dima Damen. Fine-grained action retrieval through multiple parts-of-speech embeddings. In *ICCV*, pages 450–459, 2019. 2, 3
- [16] Michael Wray, Hazel Doughty, and Dima Damen. On semantic similarity in video retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3650–3660, 2021. 1, 2, 3, 4
- [17] Yue Zhao and Philipp Krähenbühl. Training a large video model on a single machine in a day. *arXiv preprint arXiv:2309.16669*, 2023. 2, 4, 5
- [18] Yue Zhao, Ishan Misra, Philipp Krähenbühl, and Rohit Girdhar. Learning video representations from large language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6586–6597, 2023. 2, 4
- [19] Zhun Zhong, Liang Zheng, Donglin Cao, and Shaozi Li. Re-ranking person re-identification with k-reciprocal encoding. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1318–1327, 2017. 2